NFH-Kuza-Google — Unified Multi-Domain Specification

Single platform serving Local Services (Agri/Rural) and Health Services

One Flow Engine. Multiple Domains. Shared Infrastructure.

Kuza — Platform | **Google — Cloud & AI** | **Beckn — Open Network**

National Disease Programme — Health Partner

Version 2.0 | 10 April 2026 | Status: Draft

Table of Contents

1. Executive Summary
2. Unified Flow Engine — 7-Step Model
3. Actors — Generalised Roles
4. Service Categories & Taxonomy
    - Local Services Domain
    - Health Services Domain
5. Detailed Flow Specifications
    - Livelihood Services Flow (Local)
    - Household Services Flow (Local)
    - Market Produce Flow (Local)
    - Advisory Content Flow (Local)
    - TB Screening Flow (Health)
    - TB Treatment & Adherence Flow (Health)
    - TB Supervision Flow (Health)
    - Post-TB Lung Health Flow (Health)
    - Active Case Finding — CHP Outreach (Health)
    - Contact Tracing (Health, cascade)
6. Data Model
    - Farmer Profile (Local)
    - Provider Profile (Local)
    - Patient Profile (Health)
    - Health Facility Profile (Health)
    - Coordinator Profile (Health)
    - Transaction Fields — Seeker Side
    - Transaction Fields — Provider Side
    - Health Transaction Fields
7. Service Status Lifecycle (Unified)
8. Domain Detection & User Routing
9. WhatsApp Message Formats
10. Location & Matching

11. Payment Model

12. Provider Onboarding & Catalog Management

13. Integration Points

- WhatsApp Cloud API
- Beckn Network
- Google Cloud Services
- Kuza Platform
- Health-System Integrations (registers / DHIS2 / FHIR)

14. Sovereign Data & Compliance

15. Non-Functional Requirements

16. Pilot Scope & Future Extensions

17. Appendix — Phased Analytics Roadmap

1. Executive Summary

NFH-Kuza-Google is a unified service-orchestration platform that connects **seekers** (individuals, households, patients) with **providers** (agripreneurs, vets, aggregators, health facilities, clinicians) through a **single flow engine** delivered over **WhatsApp** and **web apps**, using the **Beckn open network protocol** for interoperability.

The platform now serves **two domains** in parallel, sharing the same architecture, channels, flow engine, and data infrastructure:

Domain	Beckn Code	Primary Partners	Flows
Local Services Agri / Rural livelihoods	`local-services`	Kuza, agripreneur networks, rural aggregators	Livelihood, Household, Market, Advisory
Health Services TB / National Disease Programme	`health-services`	National Disease Programme, Ministry of Health (Kenya), Global Fund, county health systems	TB Screening, TB Treatment, TB Supervision, Post-TB Lung Health

PARTNERS

Kuza — Platform technology, seeker & provider networks, field operations, domain expertise (agri + health).

Google — Cloud infrastructure (GCP) hosting the unified stack; Content APIs serving curated domain content (agri, health, market, finance); AI interfaces (location normalisation, content tailoring, triage assist, translation, generation).

Beckn — Open network protocol, DeDi Registry, ONIX adapters, Catalog & Discovery Services — one network, two domain codes.

National Disease Programme — Kenya National TB & Leprosy Programme — health-domain partner, clinical guidelines, disease-programme register alignment, Global Fund compliance.

Key Characteristics

Characteristic	Description
Unified Flow Engine	Both domains share a single 7-step flow model (Need → Search → Select → Confirm → Deliver → Record → Outcome). Domain-specific extensions plug in as schemas.
Channel-first	Seekers (farmers, patients, CHPs) interact via WhatsApp — no app installation. Providers use WhatsApp + browser-based web apps.
Multi-domain, single platform	Same webhook, same auth, same storage, same analytics stack serves both Agri and Health. Domain is a discriminator, not a separate system.
Cross-role	One person can be a seeker in one flow and a provider in another (e.g., a CHP screens patients AND receives mentorship supervision).
Location-based	Providers/facilities matched by registered service area (County + Sub-County) and capability.
Kenya pilot	47 Counties / Sub-Counties, KES currency, M-Pesa mobile money, alignment with Ministry of Health and Kenya national TB programme.

SIMPLIFIED SCHEMA MODEL

All flows — across every domain and vertical — use the **same four generic Beckn extension schemas**. Domain/vertical-specific data lives inside a free-form `verticalAttributes` object that every payload carries.

Schema	Prefix	Role
`ServiceItemAttributes`	`sia:`	Catalog item attributes (every service / product / content item)
`ServiceOfferAttributes`	`soa:`	Price / quote / offer attributes
`ServiceFulfillmentAttributes`	`sfa:`	Delivery / visit / appointment attributes
`ServiceOrderAttributes`	`sor:`	Order / referral / claim attributes

Each payload's `verticalAttributes` carries a required `vertical` discriminator (e.g., `agri-livelihood-poultry`, `household-education`, `market-livestock-products`, `advisory-agri`, `health-tb-screening`, `health-tb-treatment`, `health-tb-supervision`, `advisory-health-tb`) plus any free-form vertical-specific fields.

2. Unified Flow Engine — 7-Step Model

All interactions on the platform — across both domains — are expressed as instances of a single **7-step flow**. A "flow" is any journey where a seeker with a need is matched to a provider who can serve it and whose outcome is recorded. Whether the need is "vaccinate 200 birds" or "screen me for TB", the pipeline is the same.

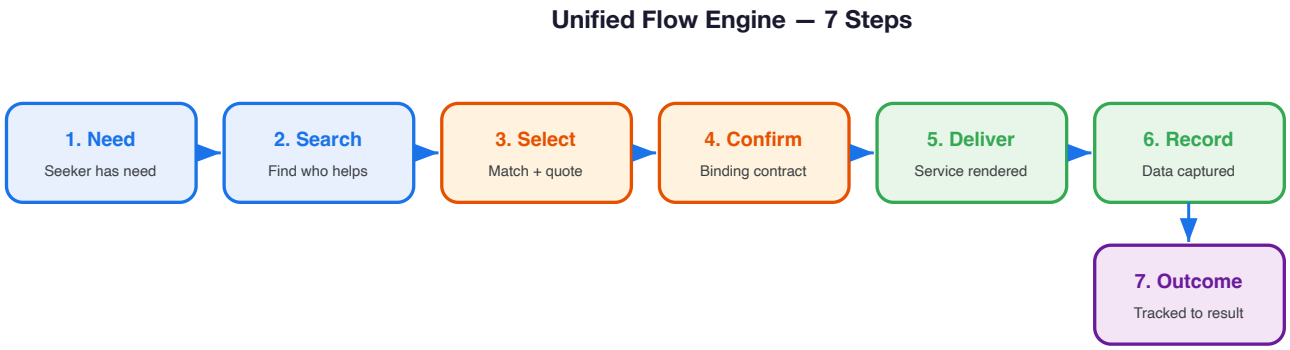

Figure 2.1 — The unified 7-step model spans both Local Services and Health Services flows.

2.1 The Seven Steps

Step	Meaning	Local Services Example	Health Services Example
1. Need	A seeker has a need	Farmer needs 200 birds vaccinated	Patient has been coughing >2 weeks; household contact needs screening
2. Search	Identify who can help, by capability + location	Discover vets serving Naivasha Sub-County	Discover screening facilities with GeneXpert serving Kisumu Central
3. Select	Match to best provider, with quote/booking	Choose vet with best rating and price	Choose nearest screening facility; generate referral slip
4. Confirm	Both sides agree — binding contract	Farmer confirms booking, provider accepts	Patient confirms appointment; facility accepts referral
5. Deliver	Service rendered	Vet visits farm and vaccinates birds	Sputum collected and sent to lab; drugs dispensed

Step	Meaning	Local Services Example	Health Services Example
6. Record	Data captured	Visit notes, amount paid, method	Clinical record, lab result, adherence log, payment claim
7. Outcome	Tracked to result	Farmer rates provider; repeat-visit flag	TB positive/negative; treatment cured / completed / failed / died / lost

2.2 Mapping to Beckn Endpoints

Flow Step	Beckn Verb(s)	Notes
Need	*(implicit)*	Captured as conversational state; not an on-network event
Search	`discover` / `on_discover`	Provider/facility catalog query
Select	`select` / `on_select` → `init` / `on_init`	Quote/booking creation
Confirm	`confirm` / `on_confirm`	Binding order / referral / appointment
Deliver	`status` updates	In-progress telemetry
Record	`update` + `status`	Clinical / operational record written back to registers
Outcome	`rate` / `on_rate` + custom outcome events	Health outcomes (cured / lost / died) are captured as structured outcome payloads

> **WHY THIS MATTERS**
>
> A single conversational engine, a single order state machine, a single analytics pipeline, and a single Beckn adapter serve both domains. Adding a new vertical (HIV, malaria, cash-transfer) becomes a schema + taxonomy exercise — not a new platform.

2.3 Flow Initiation Modes

Not every flow starts with a seeker asking the chatbot. The engine supports three initiation modes — same 7 steps, different trigger:

Mode	Who triggers	Example
`SEEKER_INITIATED`	Seeker via WhatsApp	Farmer requests vaccination; patient with cough asks for screening
`PROVIDER_INITIATED`	Provider / Coordinator proactively (batch or single)	CHP conducts Active Case Finding in a village (see §5.9); extension officer runs mass vaccination campaign
`SYSTEM_INITIATED`	Flow Engine itself — triggered by an outcome event, scheduled policy, or cascade rule	Contact tracing: positive TB diagnosis auto-creates screening flows for household/workplace/school contacts (see §5.10); outbreak quarantine triggers vaccination flows for neighbouring farms

2.4 Cascading & Linked Flows

A flow's *Outcome* step can become the *Need* step of one or more new flows. Every flow records:

- `triggerFlowId` — reference to the parent flow that spawned this one (absent for root flows)
- `triggerReason` — `CONTACT_OF_POSITIVE`, `OUTBREAK_QUARANTINE`, `SCHEDULED_FOLLOWUP`, `ROUTINE_RECALL`, etc.
- `indexEntityId` (vertical-specific) — the seeker whose outcome initiated the cascade (index patient UUID for TB contact tracing)

The parent/child linkage is preserved in the data layer, enabling surveillance (what % of contacts got screened?), yield analysis (how many new positives per index case?), and longitudinal outcome tracking.

2.5 Batch / Campaign Flows

One provider session can produce many flows. A CHP visits a village, screens 40 people — one session, 40 flows, sharing a `batchId`. The batch is a grouping for logistics, stipend payment, and reporting — each individual flow still has its own seeker, its own Record step, and its own Outcome.

- `batchId` (UUID) — groups flows into a session
- `batchType` — `CAMPAIGN`, `COMMUNITY_VISIT`, `CONTACT_TRACING_SESSION`, `OUTBREAK_RESPONSE`

Why batches matter: CHPs claim stipends per session, supervisors review batch completeness, analytics measure per-session yield.

3. Actors — Generalised Roles

Rather than hard-coding "Farmer" and "Provider", the platform models five **abstract roles**. A single individual can occupy different roles in different flows.

3.1 Seeker

An individual or household with a need that can be served by the network. A seeker may be **active** (they start the flow themselves) or **passive** (the flow is started on their behalf by a provider or by the system).

- **Farmer / Rural Household** Local — discovers and books rural services; sells produce. Usually active.
- **Patient / Family** Health — discovers TB screening, treatment, post-TB follow-up. Active when self-presenting.
- **Household contact** Health — needs preventive screening after a household TB case. Usually *passive* — the flow is triggered by the index patient's positive outcome (see §5.10 Contact Tracing).
- **Community member under ACF outreach** Health — screened during a CHP community visit. *Passive* — CHP initiates the flow, seeker consents to participate (see §5.9 ACF).
- **Primary interface:** WhatsApp for active seekers; outbound WhatsApp / SMS notification for passive seekers (informing them of a scheduled visit or requested consent).
- **Profile captured once:** name, phone, location (village/ward/county), optional GPS, preferred language, domain-specific attributes (enterprise for farmer; gender/DOB/consent for patient).

3.2 Provider

An entity that delivers the service. Providers can also *initiate* flows proactively (see Flow Initiation Modes in §2) — for example, a CHP conducting a community screening session doesn't wait for patients to walk in; the CHP starts the flow.

- **Agripreneurs / Service Providers** Local — vets, agro-vets, aggregators, trainers, tradespeople.
- **Health Facilities** Health — dispensaries, health centres, sub-county hospitals, county referral hospitals, labs (GeneXpert, culture).

- **Clinical staff** Health — clinicians, nurses, lab technicians, pharmacy staff (linked to a facility).
- **Primary interface:** WhatsApp notifications + browser-based Web App for detailed management.
- A single provider/facility can offer services across multiple categories within its domain.
- **Provider-initiated flows:** outreach screening, campaign vaccination, scheduled community visits — the provider is both trigger and fulfiller.

3.3 Coordinator

Supervisory or oversight roles that operate across many providers.

- **Extension Officers** Local — area-level agri coordination.
- **SCTLCs** Health — Sub-County TB & Leprosy Coordinators.
- **CTLCs** Health — County TB & Leprosy Coordinators.
- **National Disease Programme Director** Health — national programme leadership with oversight across all counties.
- In the platform, Coordinators appear as Seekers when consuming supervision flows, and as Validators when signing off on visits.

3.4 Validator

Labs, auditors, QA reviewers — parties whose confirmation is required to close a flow.

- **Labs** Health — return GeneXpert / culture / DST results.
- **Data-quality auditors** Health — validate facility data quality and mortality audits.
- **Agri-input auditors** Local — optional QA on input provenance.

3.5 Sponsor

Entities that fund, commission, or benefit from aggregate outcomes.

- Ministry of Health, Global Fund, TaifaCare Health
- Donors, NGOs, government programmes Local Health

3.6 Cross-Cutting Role — CHV / CHP

Community Health Volunteers / Community Health Promoters are particularly important because they simultaneously play **multiple roles**:

- As a **Provider who initiates flows**: CHPs proactively visit communities, households, schools, or workplaces and conduct screening sessions. Each visit is a provider-initiated batch of individual flows (see §5.9 Active Case Finding).
- As a **Provider who responds to cascades**: when a patient tests positive, the Flow Engine creates screening flows for contacts; CHPs are typically assigned to visit those contacts and conduct the screening (see §5.10 Contact Tracing).
- As a **Seeker**: they need supervision, payment claims, mentorship, training.
- The platform supports this by allowing a single user record to be attached to multiple role-profiles; context (which flow, which domain) determines which role is active.

4. Service Categories & Taxonomy

Each domain exposes its own top-level categories. Sub-categories are dynamic master data that can be extended without schema changes.

4.1 Local Services Domain `local-services`

Category	Example Sub-Categories	Example Services
Livelihood-Services	Livestock Services, Crop Services, Trades & Skills	Vaccination, brooding, deworming, feed supply, ploughing, spraying, tailoring, welding, carpentry
Household-Services	Education, Health, Insurance, Consumer Goods	School fees, tutoring, books, hospital checkup, pharmacy, crop/health insurance, household items
Market-Produce	Farm Produce, Livestock Products, Handicrafts & Goods	Vegetables, grains, fruits, eggs, milk, meat, live birds, handmade goods
Advisory-Content	Agricultural Advisory, Financial Literacy, Health & Nutrition, Business Skills	Poultry care, crop management, budgeting, nutrition guides

4.2 Health Services Domain `health-services`

Category	Example Sub-Categories	Example Services
TB-Screening	Symptom screening, Sputum collection, GeneXpert referral, Chest X-ray, Contact tracing, Active Case Finding (ACF)	Household contact screening, CHP-led symptom screening, GeneXpert test referral, CXR mobile clinic
TB-Treatment	DSTB, DRTB, TPT (TB Preventive Therapy), Pediatric TB, Adherence monitoring, ADR reporting	Monthly drug dispensing, adherence reminder, ADR capture, sputum conversion check
TB-Supervision	Facility supervision, CHP supervision, Data quality audit, Mortality audit, Mentorship	SCTLC quarterly facility visit, CHP household review, death audit, mentorship session
Post-TB-Care	PTLD (Post-TB Lung Disease) screening, Spirometry, Pulse oximetry, Lung function follow-up, Asthma management	Post-cure spirometry, 6-month PTLD review, referral to pulmonology

DYNAMIC TAXONOMY

Sub-categories in both domains are **not hardcoded**. They are managed as master data in Kuza Platform and pulled dynamically. New sub-categories — or new top-level categories in additional verticals (HIV, malaria, NCDs) — can be added without schema changes.

5. Detailed Flow Specifications

5.1 Livelihood Services Flow Local

Example: Poultry Vaccination in Nakuru County

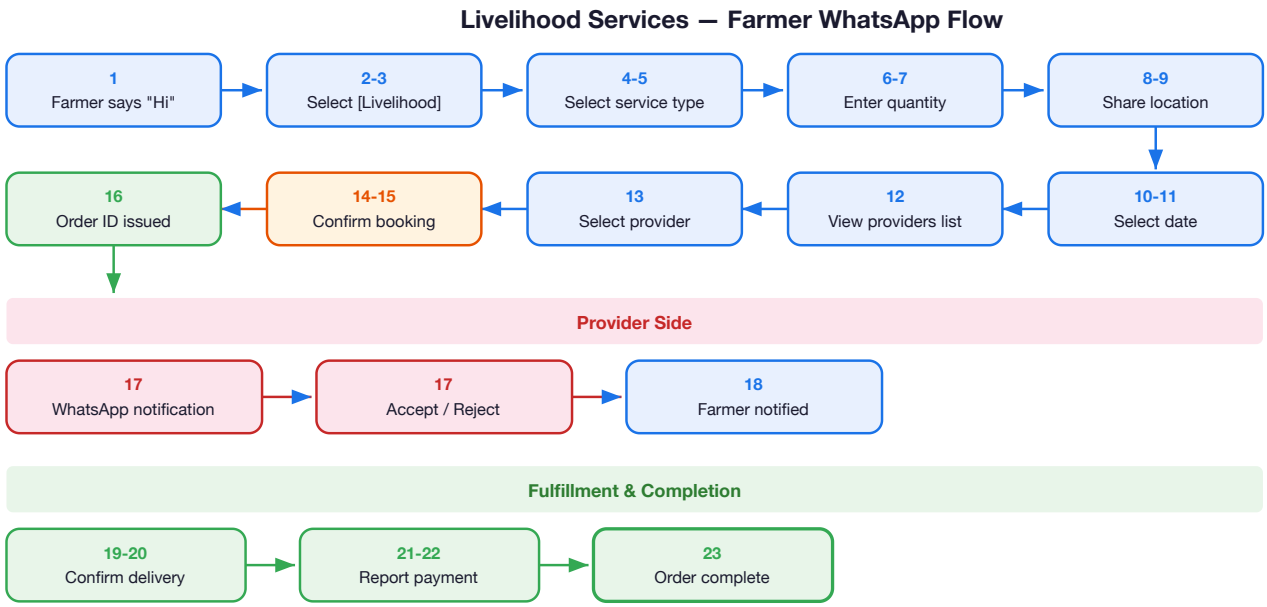

Figure 5.1 — Livelihood Services: End-to-end flow from farmer request to completion

Farmer WhatsApp Flow (Step by Step)

1. Farmer sends **"Hi"**
2. System: "*Welcome. What do you need help with?*" — 4 buttons: Livelihood, Household, Market, Advisory
3. Farmer selects **[Livelihood]**
4. System: "*What service do you need?*" — Numbered list: 1. Vaccination 2. Inputs 3. Brooding 4. Incubation (pulled from master data)

5. Farmer selects "1" (Vaccination)
6. System: "*How many birds do you need vaccinated?*"
7. Farmer enters number (e.g., "200")
8. System: "*Please share your location*" — farmer shares village/ward/county or live GPS location
9. Google AI normalises location. System confirms: "*Nakuru County, Naivasha Sub-County. Correct?*" [Yes] [No]
10. System: "*When do you need the service?*" — buttons: Today, Tomorrow, Choose date
11. Farmer selects preferred date
12. System: "*Available service providers near you:*" — Numbered list showing provider name, price, rating
    `1. John Kariuki - KES 30/bird - 4.5 stars`
    `2. Peter Omondi - KES 25/bird - 4.2 stars`
13. Farmer selects provider (e.g., "1")
14. System shows booking summary: Service, Date, Provider, Total Cost (200 birds x KES 30 = KES 6,000) — [Confirm] [Cancel]
15. Farmer taps **[Confirm]**
16. System: "*Your request has been sent to the provider. Order ID: ORD-20260412-001*"
17. Provider receives WhatsApp notification with Accept/Reject buttons
18. System updates farmer: "*Your request has been accepted by John Kariuki. Visit on 12 Apr 2026. Contact: +254 712 345 678*"
19. After service, System asks: "*Was the vaccination completed?*" [Yes] [No]
20. Farmer taps **[Yes]**
21. System asks: "*How did you pay?*" — buttons: [Cash] [M-Pesa]
22. Farmer selects payment method
23. System: "*Thank you. Your request is completed.*"

Provider Flow for Same Order

1. Provider receives WhatsApp: "*New order received! Vaccination - 200 birds. Farmer: Mary Wanjiku, Naivasha, Nakuru. Date: 12 Apr 2026. Amount: KES 6,000*" — [Accept] [Reject] [View Details]
2. Provider taps **[Accept]** → order status changes to `PROVIDER_ACCEPTED` → farmer notified
3. Provider taps **[View Details]** → opens Kuza Web App showing full order

4. After completing service, provider opens web app → marks complete → enters: actual service date, payment received (yes/no), amount, payment method, notes, follow-up required (yes/no)
5. System triggers farmer confirmation flow (steps 19–23 above)

Beckn API Mapping

Flow Step	Beckn Action	Direction	Payload Summary
Steps 3–11 (category + service + location + date)	`discover`	BAP → Network	category, subCategory, quantityRequired, farmerLocation, scheduledDate
Step 12 (provider list)	`on_discover`	Network → BAP	Catalog with providers, items, offers filtered by service area
Step 13 (select provider)	`select`	BAP → BPP	Selected provider ID + item ID + offer ID
Quote confirmation	`on_select`	BPP → BAP	Price quote, cancellation policy, provider details
Steps 14–15 (confirm booking)	`init` + `confirm`	BAP → BPP	Farmer contact, scheduled date, quantity, confirmation
Step 16 (order created)	`on_confirm`	BPP → BAP	Order ID, provider details, status=CONFIRMED
Steps 17–18 (provider accepts)	`status` update	BPP → BAP	status=CONFIRMED, confirmedVisitTime
Steps 19–22 (completion + payment)	`status` + `update`	Both	status=DELIVERED → RECORDED, payment details
Feedback	`rate` / `on_rate`	Both	Outcome objects

5.2 Household Services Flow Local

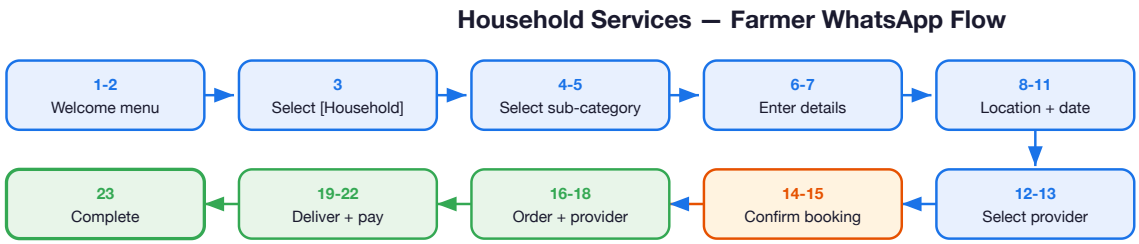

Figure 5.2 — Household Services: Structurally identical to Livelihood with domain-specific details

Structurally identical to Livelihood. Key differences:

- **Sub-categories:** Education, Health, Insurance, Consumer Goods
- **Step 6 equivalent:** Quantity/details question varies by service type (e.g., "Which school?" for fees, "How many items?" for products)
- `serviceMode` may be `FACILITY_VISIT`, `PRODUCT_DELIVERY`, `SERVICE_DELIVERY`
- `beneficiaryType` can indicate CHILD, ELDERLY, etc.

5.3 Market Produce Flow Local (Reversed Economics)

> **KEY DIFFERENCE**
>
> In this flow the farmer is the **seller** and the provider is the **buyer/aggregator**. Money flows from provider to farmer.

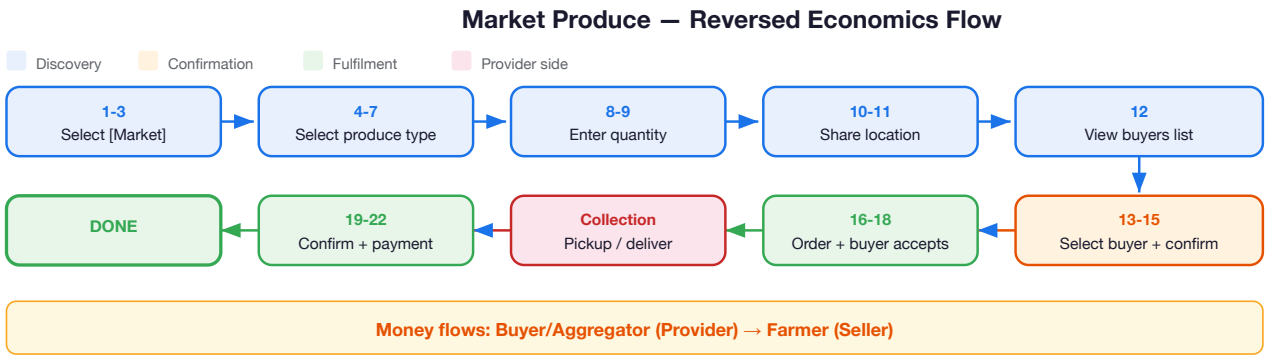

Figure 5.3 — Market Produce: Farmer is the seller, provider is the buyer. Money direction is reversed.

Key Schema Differences

- `discover` intent includes `produceCategory`, `quantityAvailable`
- `on_discover` returns aggregators with **buying prices**, `collectionMode`, `minimumQuantity`
- `collectionMode`: `FARM_GATE_PICKUP` | `COLLECTION_POINT` | `FARMER_DELIVERS`

5.4 Advisory Content Flow Local (Discover-Only)

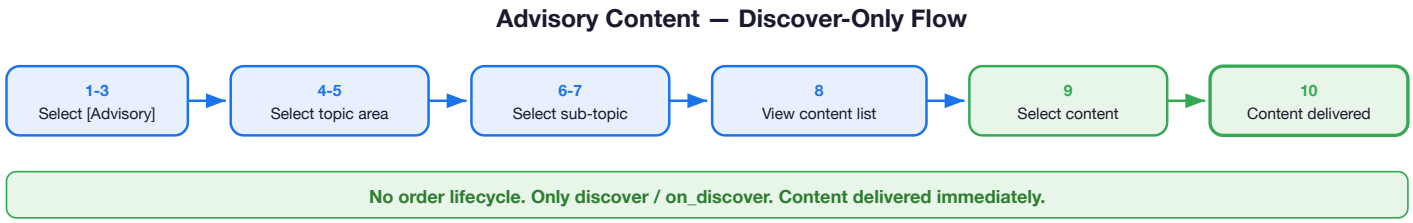

Figure 5.4 — Advisory Content: Discover-only, no transaction lifecycle

Beckn mapping: `discover` / `on_discover`. The BPP calls Google Content APIs during `on_discover` processing and optionally tailors output through Google AI Interfaces before returning the composed advisory.

5.5 TB Screening Flow Health

Example: Joseph in Kisumu has been coughing for 3 weeks.

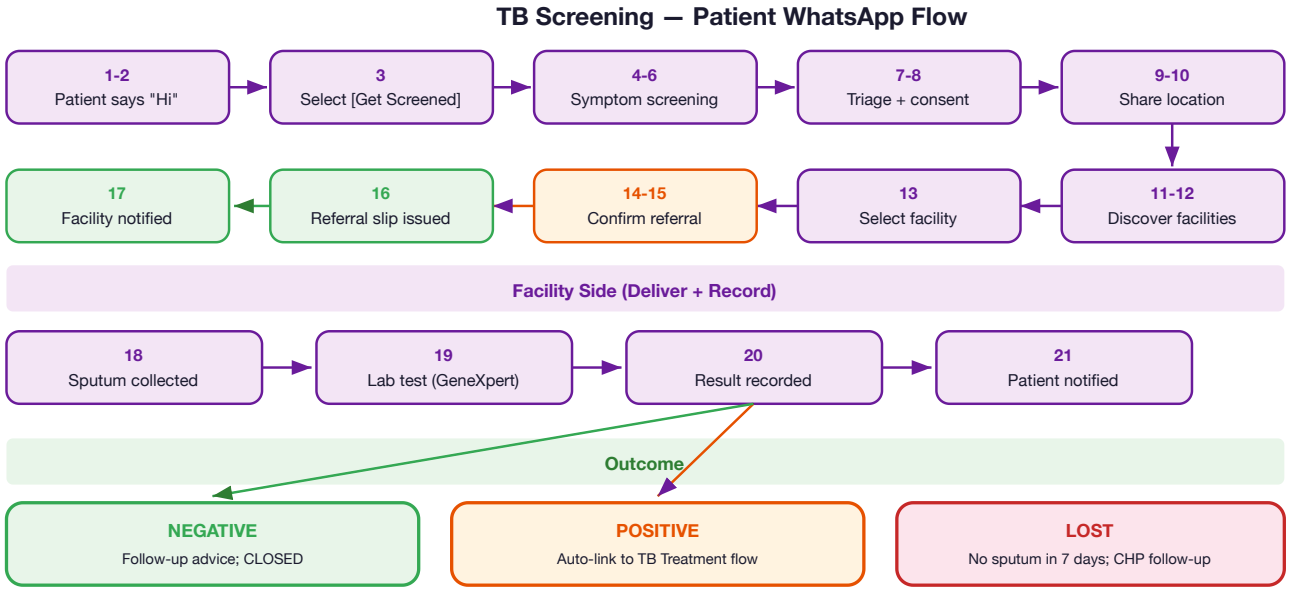

Figure 5.5 — TB Screening: Symptom → referral → sputum → lab → outcome. Positive auto-links to treatment.

Patient WhatsApp Flow (Step by Step)

1. Joseph sends **"Hi"**
2. System detects role (or asks if new): "*Welcome. How can we help?*" — Buttons: [Get Screened] [Treatment Help] [Find Facility] [Health Info]
3. Joseph taps **[Get Screened]**
4. System: "*We'll ask a few quick questions. Have you been coughing for more than 2 weeks?*" [Yes] [No]
5. Joseph: [Yes] → further symptoms (fever, weight loss, night sweats, contact with TB patient)
6. System scores symptoms (presumptive TB if any positive)
7. System: "*You may need a TB test. May we share your information with a screening facility?*" [Yes, I consent] [No]
8. Joseph consents (DPDP-style consent captured)
9. System: "*Please share your location*"

10. Google AI normalises → confirms County + Sub-County
11. System calls `discover` for facilities with capability `GeneXpert` in that location
12. System: "*Nearest facilities:*"
    `1. Kisumu County Hospital (GeneXpert, 2.3 km)`
    `2. Nyalenda Health Centre (Smear + referral, 4.1 km)`
13. Joseph selects "1"
14. System shows referral summary (facility, service, date, patient ID) — [Confirm] [Cancel]
15. Joseph taps **[Confirm]**
16. System: "*Referral sent. Slip ID: REF-20260412-017. Show this at facility.*"
17. Facility receives WhatsApp notification; clinician sees queued referral in web app
18. Joseph visits facility; sputum collected (Deliver)
19. Sample sent to GeneXpert; Validator (lab) returns result
20. Clinician records result in web app (Record) — writes back to the disease-programme register via patient UUID
21. System notifies Joseph: "*Your test result is ready. Please call the clinic.*" (clinical result delivered by clinician, not bot)
22. Outcome: If POSITIVE → system auto-creates a TB Treatment flow linked to Joseph's UUID. If NEGATIVE → follow-up advice, flow CLOSED. If patient does not appear within 7 days → flow marked LOST and CHP receives follow-up task.

Schema Example — `discover` Catalog Item

The TB screening referral uses the generic `ServiceItemAttributes` schema; the vertical-specific data lives in `verticalAttributes`:

```
{
  "id": "svc_tb_screening_geneexpert",
  "descriptor": { "name": "GeneXpert TB screening" },
  "sia:ServiceItemAttributes": {
    "serviceCategory": "TB-Screening",
    "capabilityRequired": "GeneXpert"
  },
  "verticalAttributes": {
    "vertical": "health-tb-screening",
    "screeningType": "SPUTUM_COLLECTION",
    "diseaseTarget": "Tuberculosis",
```

```
    "diagnosticMethod": "GeneXpert"
  }
}
```

5.6 TB Treatment & Adherence Flow Health

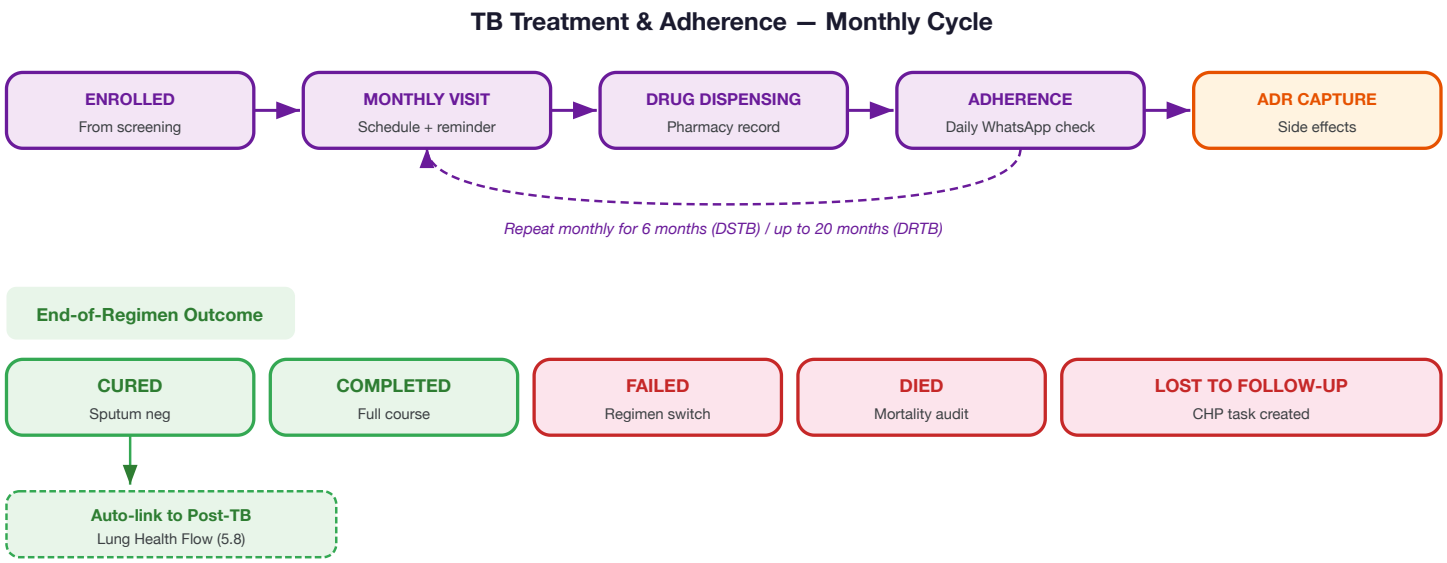

Figure 5.6 — TB Treatment: Monthly cycle repeats for the full regimen; outcome determines next-step flow.

Flow Summary

- **Need:** Patient enrolled in treatment (usually auto-linked from TB Screening positive outcome).
- **Search / Select / Confirm:** Monthly visit scheduled at the treatment facility; patient confirms via WhatsApp.
- **Deliver:** Drug dispensing at pharmacy; sputum conversion check at 2 months; daily adherence reminder via WhatsApp template.
- **Record:** Each monthly visit generates a record in disease-programme registers (e.g., DSTB / DRTB / TPT for TB) via patient UUID. ADRs captured through a structured WhatsApp form.
- **Outcome:** At end of regimen — `CURED`, `COMPLETED`, `FAILED`, `DIED`, `LOST_TO_FOLLOWUP`. `CURED` auto-creates a Post-TB Lung Health flow at the 6-month follow-up mark.

5.7 TB Supervision Flow Health

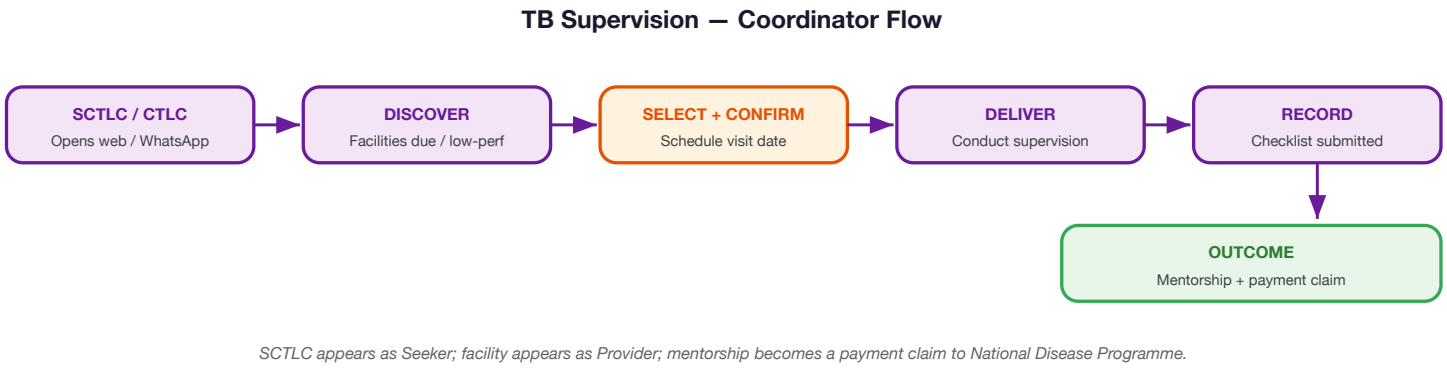

Figure 5.7 — TB Supervision: the same 7-step engine with Coordinator as Seeker and Facility as Provider.

5.8 Post-TB Lung Health Flow Health

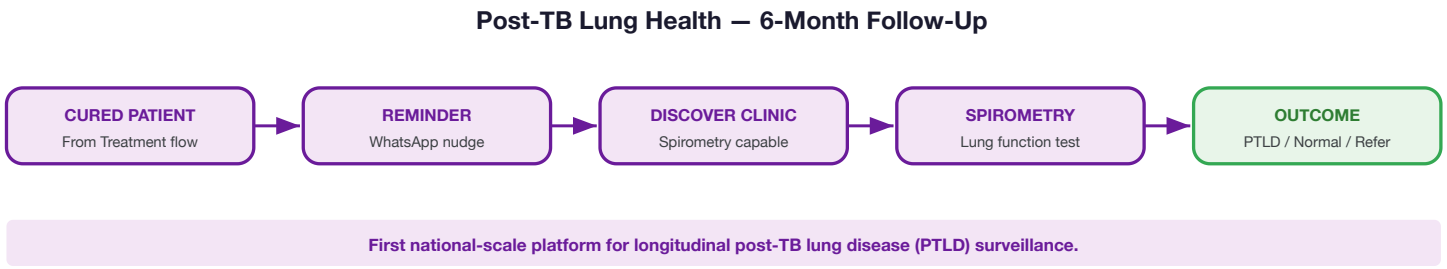

Figure 5.8 — Post-TB Lung Health: cured TB patients routed into long-term lung function surveillance.

5.9 Active Case Finding — CHP Community Outreach Health

Not every health flow begins with a patient walking in. **Active Case Finding (ACF)** is provider-initiated: a Community Health Promoter (CHP) proactively visits communities, households, schools, or workplaces to screen people for TB. The same flow engine orchestrates this — with `initiationMode = PROVIDER_INITIATED` and a shared `batchId` grouping the session's screenings.

> **Why ACF matters:** The WHO estimates that roughly one third of TB cases worldwide go undiagnosed each year because symptomatic patients never present to a facility. ACF closes that gap. Evidence from Kenya's National TB Programme shows ACF finds more than twice as many TB cases per CHP-hour compared with passive screening at facilities.

5.9.1 Example Scenario

CHP Shwetha is assigned to Katolo village in Nyakach Sub-County, Kisumu County. Today she conducts a routine community-screening visit.

1. Shwetha opens the CHP app (launched from a WhatsApp link). App pulls her assigned list for today.
2. App prompts: "*Start community screening session for Katolo village?*" [Start]
3. Flow Engine creates a **batch** (`batchType: COMMUNITY_VISIT` , `batchId: b4f1c8d6-2025-11-15-nyakach-chp-visit`).
4. Shwetha visits each household. For every person she screens, the app creates a new flow (`initiationMode: PROVIDER_INITIATED` , linked to the batch).
5. Each flow captures: person's identity (with consent), screening symptoms (cough duration, weight loss, fever, night sweats), CHP's clinical judgement, and referral decision.
6. For symptomatic persons, the flow's `outcomeType` is `SYMPTOMATIC_REFERRED` with `followUpRequired = true` ; the app can schedule a sputum-collection appointment and optionally trigger a new flow (`health-tb-screening` vertical) linked via `triggerFlowId` .
7. At end of day, Shwetha closes the batch. All 40 individual flows are in `RECORDED` or `OUTCOME_RECORDED` state. Her supervisor (SCTLC) can review the batch-level dashboard: 40 screenings, 4 symptomatic referrals, 2 sputum appointments scheduled.

5.9.2 Key Flow Attributes

Attribute	Value for ACF
`initiationMode`	`PROVIDER_INITIATED`
`batchType`	`COMMUNITY_VISIT`
`verticalAttributes.vertical`	`health-tb-acf`
Seeker (per flow)	Person being screened; role is **passive participant** — they didn't initiate, but they consent to be screened
Provider	CHP conducting the screening (with `providerCredentialId`)
Cascading	Positive screens can spawn a follow-on `health-tb-screening` flow via `triggerFlowId`

5.9.3 Operational Value

- **CHP stipends** paid per session (batch), not per individual screening — the batch abstraction enables this directly.
- **Supervision**: SCTLC reviews batch completeness and positive-rate; high positive rates may trigger a hotspot response.
- **Analytics**: positive-yield per CHP / per village / per season informs future outreach planning.

5.10 Contact Tracing — System-Initiated Cascade Health

Tuberculosis is airborne and communicable. When a patient tests positive, everyone in prolonged close contact with them during the infectious period must also be screened. The Flow Engine automates this: a positive outcome on one flow triggers a fan-out of new screening flows, one per identified contact, each linked via `triggerFlowId` and marked `initiationMode = SYSTEM_INITIATED`.

5.10.1 Contact Identification

During the index patient's interview (part of `RECORD` step), the clinician / CHP enumerates contacts:

- **Household contacts** — everyone living in the same residence
- **Workplace colleagues** — for patients in enclosed workplaces (offices, factories, transport)
- **School classmates and teachers** — for student or teacher patients
- **Social / close friends** — individuals reported as spending significant time with the patient
- **Neighbours** — for densely populated informal settlements

5.10.2 Cascade Mechanism

```
Index Flow F1 (SEEKER_INITIATED or PROVIDER_INITIATED)
    Patient Mary presents with cough. Screened, tested, diagnosed.
    outcomeType: TB_POSITIVE

Flow Engine rule fires: "positive TB diagnosis triggers contact tracing"

For each contact identified during F1.RECORD step:
    New Flow Fi created with:
        initiationMode: SYSTEM_INITIATED
        triggerFlowId: F1
        triggerReason: CONTACT_OF_POSITIVE
        batchId: (shared across all contacts of this index case)
        batchType: CONTACT_TRACING_SESSION
```

```
        verticalAttributes.vertical: health-tb-contact-tracing
        verticalAttributes.contactRelationship: HOUSEHOLD | COWORKER | CLASSMATE |
...
        verticalAttributes.indexPatientUuid: (Mary's UUID)
        verticalAttributes.exposureWindow: { startDate, endDate }

Each Fi then progresses independently through:
    Need (auto-registered) → Search → Select (CHP home visit) → Confirm → Deliver → R
ecord → Outcome
```

5.10.3 Outbound Notifications

Contact-traced individuals are **passive participants** — they didn't start the flow. The Flow Engine proactively reaches them via WhatsApp (or SMS if WhatsApp unavailable):

> **Example WhatsApp notification to contact:**
>
> ```
> Habari. You have been identified as a contact of someone recently diagnosed w
> ith TB. Don't worry — TB is treatable, and early screening protects you and y
> our family.
>
> A Community Health Promoter will visit your home on [date] at [time] for a fr
> ee screening. Reply [YES] to confirm, or [RESCHEDULE] if the time does not wo
> rk.
>
> This is a private, confidential communication. We do not share the name of th
> e index patient.
> ```

The notification is non-identifying (never names the index patient) and respects consent. If the contact declines or cannot be reached, the flow transitions to `LOST_TO_FOLLOWUP` for surveillance reporting.

5.10.4 Clinical and Ethical Safeguards

- **Consent required**: index patient must consent to share contacts; each contact must consent to screening. Both recorded with timestamps.
- **No identifying disclosure**: index patient's identity is never revealed to contacts.

- **Data minimisation**: only contact-tracing-relevant data flows to the cascaded flows (exposure window, relationship type, geographic overlap) — not the index patient's full clinical record.
- **Stigma mitigation**: outbound messaging is framed positively (treatable, free, protective) per WHO guidance.

5.10.5 Analytics Value

Linkage (`triggerFlowId` chain) enables surveillance metrics that are otherwise impossible:

- Contact-investigation completeness: what % of identified contacts of a positive case are actually screened within 2 weeks?
- Yield: how many new cases are found per index case, by relationship type (household vs workplace vs school)?
- Cluster detection: are positive cases clustering around specific index patients, locations, or time windows?
- Longitudinal impact: post-TB outcomes for the cohort of index+contacts over months / years.

6. Data Model

6.1 Farmer Profile Local

Field	Type	Source	Required
`farmer_name`	string	WhatsApp registration	Yes
`phone_number`	string	Auto from WhatsApp (`wa_id`)	Yes
`village` / `ward` / `county` / `sub_county`	string	Captured once	County+Sub-County required
`gps_latitude` / `gps_longitude`	decimal	Optional live location	No
`primary_enterprise`	string	Optional	No
`preferred_language`	string	ISO 639-1	No

6.2 Provider Profile Local

Field	Type	Source	Required
`provider_name`	string	Registration / onboarding	Yes
`phone_number`	string	Auto from WhatsApp	Yes
`provider_type`	string	vet, agrovet, aggregator, trainer, etc.	Yes
`services_offered`	relation	Links to catalog items	Yes
`price_list`	relation	Links to offers per service	Yes
`service_areas`	relation	Counties + Sub-Counties served	Yes
`capacity` / `availability`	object	Per-service	No

6.3 Patient Profile Health (One-time Fields)

Field	Type	Source	Required
`patient_uuid`	string (UUID)	System-generated; links to disease-programme registers	Yes
`name`	string	WhatsApp onboarding	Yes
`phone_number`	string	Auto from WhatsApp	Yes
`gender`	enum (M/F/Other)	Onboarding	Yes
`date_of_birth`	date	Onboarding	Yes
`village` / `ward` / `sub_county` / `county`	string	Captured once	County+Sub-County required
`next_of_kin`	object {name, phone, relation}	Onboarding	Recommended
`consent_for_data_sharing`	boolean + timestamp	Explicit DPDP-style consent	Yes
`preferred_language`	string	ISO 639-1	No

6.4 Health Facility Profile Health

Field	Type	Source	Required
`facility_id`	string	Links to the health-facility register	Yes
`facility_name`	string	Registration	Yes
`facility_type`	enum (`DISPENSARY` | `HEALTH_CENTRE` | `HOSPITAL` | `LAB`)	Registration	Yes
`location`	object {county, sub_county, ward, gps}	Registration	Yes
`capabilities`	array of strings	e.g. [`GeneXpert` , `X-ray` , `Spirometry` , `DRTB-certified`]	Yes
`staff_credentials`	array of objects	Credentialed clinical staff	Recommended
`service_hours`	object	Weekly schedule	Recommended

6.5 Coordinator Profile Health (SCTLC / CTLC)

Field	Type	Notes
`coordinator_id`	string	Unique identifier
`name`	string	
`level`	enum (`SUB_COUNTY` | `COUNTY` | `NATIONAL`)	
`assigned_geography`	object	Counties / Sub-Counties under oversight
`assigned_facilities`	array of `facility_id`	Facilities under supervision

6.6 Transaction Fields — Seeker Side (all domains)

Field	Type	Captured When
`service_needed`	string	Search step
`quantity / details`	varies	Need elaboration
`preferred_date`	date	Search step
`selected_provider`	relation	Select step
`booking_confirmed`	boolean	Confirm step
`order_id`	string	System generated on confirm
`order_status`	enum (see § 7)	Throughout lifecycle
`service_confirmed`	boolean	Deliver step
`payment_method`	enum (`CASH` | `MPESA`)	Record step (local)
`payment_status`	enum	Record step

6.7 Transaction Fields — Provider Side (all domains)

Field	Type	Captured When
`request_id`	string	On notification

Field	Type	Captured When
`service_requested`	string	From order
`seeker_location`	object	From order
`acceptance_status`	enum	Select/Confirm response
`confirmed_visit_time`	datetime	Confirm
`fulfillment_status`	enum	After Deliver
`notes`	string	Record
`follow_up_required`	boolean	Record

6.8 Health Transaction Fields Health (per request)

Field	Type	Notes
`transaction_id`	string	Unique per interaction
`flow_template`	enum	e.g. `tb-screening` , `tb-treatment` , `tb-supervision` , `post-tb`
`vertical`	string	e.g. `tb` — allows extension to HIV, malaria, etc.
`patient_uuid`	string	Links to disease-programme register
`service_status`	enum	Uses unified flow-engine lifecycle (§ 7)
`outcome_type`	enum	`TB_POSITIVE` | `TB_NEGATIVE` | `CURED` | `COMPLETED` | `FAILED` | `DIED` | `LOST_TO_FOLLOWUP` | `PTLD_DETECTED` | `NORMAL_LUNG`
`adverse_events`	array of objects	ADR captures (MedDRA-aligned where possible)
`referral_target`	string	Next facility for escalated care

7. Service Status Lifecycle (Unified)

The lifecycle aligns 1:1 with the 7-step Flow Engine. Every flow — from vaccination bookings to TB treatment — moves through the same states.

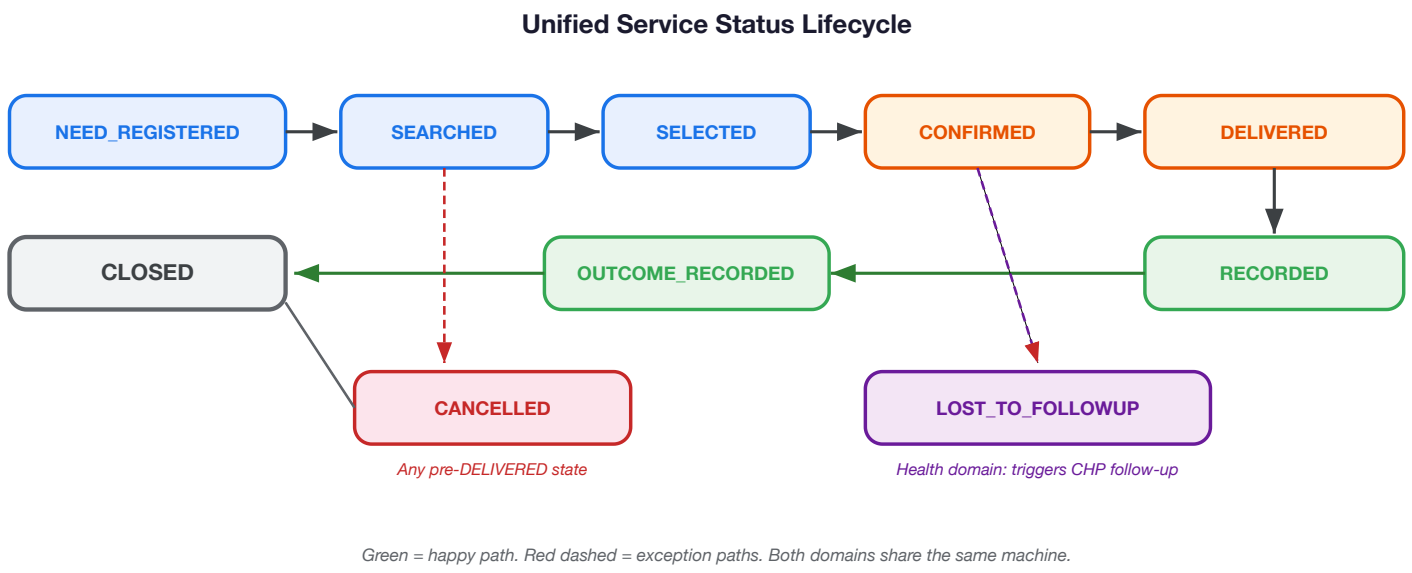

Figure 7.1 — Unified service status lifecycle aligned to the 7-step Flow Engine.

State Definitions

State	Description	Maps to Flow Step
`NEED_REGISTERED`	Seeker initiates a flow; need captured	1. Need
`SEARCHED`	Discovery completed; candidates returned	2. Search
`SELECTED`	Seeker picked a provider/facility; quote issued	3. Select
`CONFIRMED`	Both sides agree — binding contract	4. Confirm
`DELIVERED`	Service rendered	5. Deliver
`RECORDED`	Clinical / operational data written	6. Record
`OUTCOME_RECORDED`	Final outcome logged (rating, cure, etc.)	7. Outcome

State	Description	Maps to Flow Step
`CLOSED`	Terminal state	—
`CANCELLED`	Either party cancelled before DELIVERED	—
`LOST_TO_FOLLOWUP` Health	Patient did not return for scheduled step	—

Backward Compatibility — Local Services State Mapping

Existing Local Services implementations that use the legacy state names continue to work via this mapping:

Legacy (Local)	Unified
`ORDER_PLACED`	`CONFIRMED`
`PROVIDER_ACCEPTED`	`CONFIRMED` (with provider-ack sub-status)
`PROVIDER_REJECTED`	`CANCELLED`
`IN_PROGRESS`	`DELIVERED` (sub-status: in-progress)
`COMPLETED`	`DELIVERED`
`FARMER_CONFIRMED`	`OUTCOME_RECORDED`

8. Domain Detection & User Routing

Both domains share a single WhatsApp entry point but route to domain-specific handlers. The platform determines which domain a seeker belongs to and which menu to show.

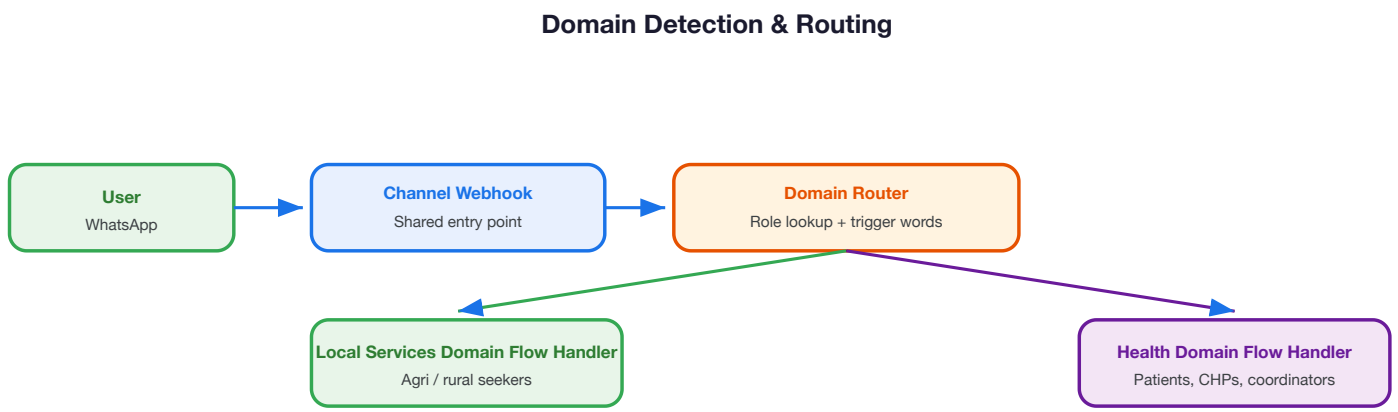

Figure 8.1 — A single webhook; a lightweight router dispatches to the correct domain handler.

Routing Strategy

1. **Role lookup:** If the WhatsApp `wa_id` is already registered with a domain, route directly to that handler. A user can be registered with more than one domain; in that case, the last-used domain is default.
2. **Trigger words:** First-time or ambiguous messages can carry a trigger — "*screen me*", "*cough*", "*TB*" → health; "*vaccine*", "*eggs*", "*school fees*" → local.
3. **Menu-driven disambiguation:** If role and triggers are absent, the system shows a top-level domain menu: "*Are you looking for: (a) Rural services (b) Health services*".
4. **Cross-domain re-entry:** A user can switch domains mid-conversation by typing "*menu*" or via a persistent footer button.

Example — Two Users, Two Domains

> **Farmer Mary** sends "Hi" → router finds `role=FARMER` → Local Services Domain Flow Handler

→ "*Welcome. What do you need help with?*" [Livelihood] [Household] [Market] [Advisory]

Patient Joseph sends "Hi" → router finds `role=PATIENT` → Health Domain Flow Handler → "*Welcome. How can we help?*" [Get Screened] [Treatment Help] [Find Facility] [Health Info]

9. WhatsApp Message Formats

Format	When Used	Constraints
Interactive Buttons	Binary or small choices (Yes/No, Confirm/Cancel, top-level domain menu)	Max 3 buttons, 20 chars each
Numbered Text List	Lists with more than 3 items: sub-categories, providers, facilities	`1. Option A\n2. Option B` . User replies with number.
Free Text Input	Quantities, locations, symptoms, feedback	Validated by engine; re-prompt on invalid input
Media Messages	Advisory content, patient education videos, referral slip PDFs	Supported: MP4, MP3, PDF, JPEG/PNG
Template Messages	Outbound notifications, adherence reminders, supervision alerts	Pre-approved WhatsApp Business templates; required parameters
Structured Forms	ADR capture, symptom checklists, supervision checklists	WhatsApp Flows (form UI); responses normalised server-side

10. Location & Matching

Location Capture

- Seeker provides location as **free text** (village/ward/county) or **GPS live location**.
- **Google AI normalises** free text to structured `County + Sub-County`.
- System confirms normalised location with seeker before proceeding.

Provider / Facility Matching

- Providers register **explicit service areas** (County + Sub-County combinations).
- Health facilities additionally have **capability filters** (e.g., `GeneXpert`, `Spirometry`, `DRTB-certified`).
- **Match logic:** seeker's `county + sub_county` must exist in provider's registered service areas AND the provider must advertise the required capability.

Kenya Geography

- **47 Counties**, each with multiple Sub-Counties.
- Pilot will start with selected counties and expand.

11. Payment Model

PILOT CONSTRAINT

All seeker payments are **offline and self-reported**. There is no online payment processing in the pilot.

Aspect	Local Services	Health Services
Payment methods	Cash, M-Pesa	Public services are free to the patient; supervision mentorship and CHP stipends are paid via M-Pesa claims to National Disease Programme
Direction	Seeker → Provider (Market flow reversed)	Sponsor (National Disease Programme / Global Fund) → Facility / CHP (claims-based)
Reporting	Self-reported by both parties	Structured claim events written to Record step; anomaly detection in Phase 1 analytics (see Appendix)
Reconciliation	No automated reconciliation in pilot	Claim-verification workflow tied to Coordinator role

12. Provider Onboarding & Catalog Management

Existing Agripreneurs Local

Already in Kuza Platform `user` table; supplement with service offerings, pricing, service areas.

New Provider Onboarding Local

1. **Registration** — name, phone, provider type
2. **Service area setup** — Counties + Sub-Counties
3. **Catalog creation** — category, sub-category, pricing, capacity
4. **Price list management**

Health Facility Onboarding Health

1. **Link to health-facility register** — existing national facility record (e.g., Kenya MoH register) is the source of truth.
2. **Capability registration** — GeneXpert, X-ray, Spirometry, DRTB, etc.
3. **Staff onboarding** — credentialed clinical staff linked to the facility.
4. **Coordinator assignment** — each facility has an SCTLC / CTLC responsible for supervision.
5. **Service hours & contact**.

All management through the **Kuza Web App**. Catalog published to the Beckn network via `BPP → ONIX → Catalog Service`.

13. Integration Points

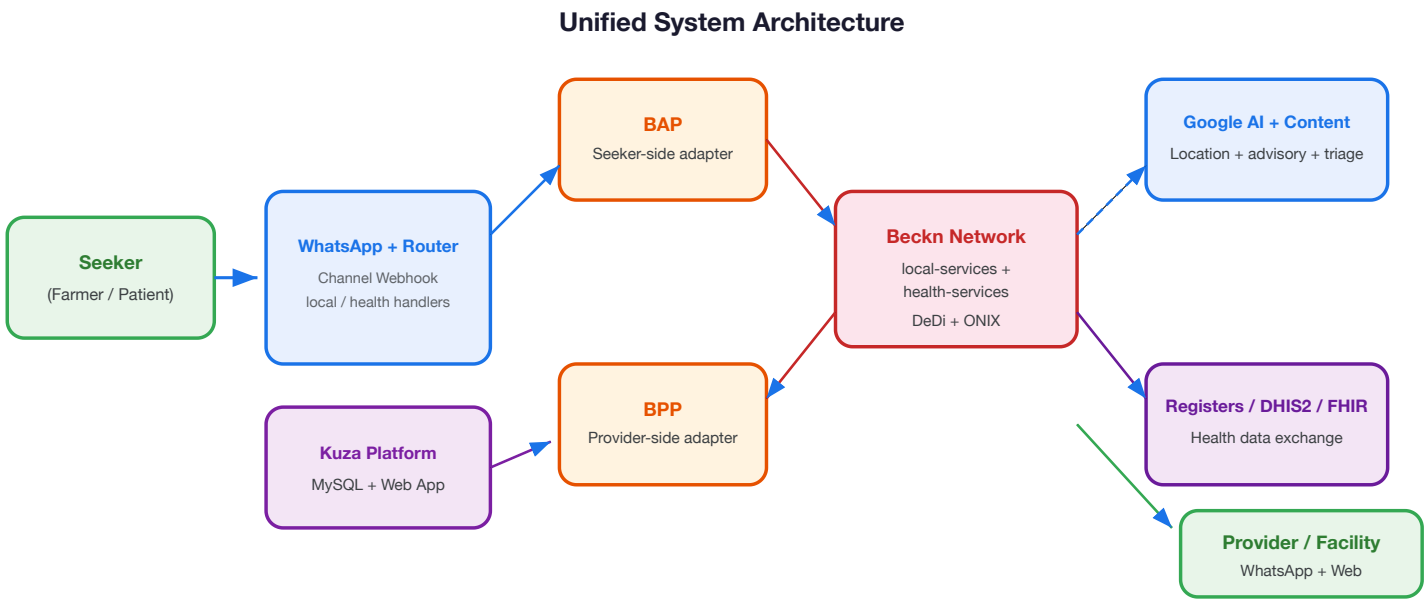

Figure 13.1 — Unified architecture: one stack, two Beckn domains, pluggable integrations.

13.1 WhatsApp Cloud API

Direction	Mechanism	Details
Inbound	Channel Webhook Endpoint	Receives all WhatsApp messages; router dispatches to Local Services Domain Flow Handler or Health Domain Flow Handler
Outbound to seeker	Domain handlers	Interactive messages, text, media, forms
Outbound to provider/facility	Messaging queue / cron	Pre-approved template messages with action buttons

13.2 Beckn Network

- **BAP & BPP** registered with DeDi Registry, each handling **both** `local-services` and `health-services` domain codes.
- **ONIX adapters** deployed on GCP; schema extensions per domain.
- Catalog and Discovery Services are shared; domain acts as a filter.

- Transactions (`select` / `init` / `confirm` / `status` / `update` / `rate`) routed directly BAP ↔ BPP via DeDi Registry.

13.3 Google Cloud Services

13.3.1 Google Cloud Platform (GCP)

- Hosts the entire Kuza-built stack: WhatsApp Handler, BAP, BPP, Web App, ONIX adapters, MySQL database.
- Networking, TLS termination, load balancing, public endpoints for BAP / BPP callbacks.

13.3.2 Google Content APIs

- **Domains:** agri, health, market, finance, education — each with its own curated knowledge base.
- **Primary consumer:** Advisory flow (local) and Health Info flow (health).
- **Secondary consumers:** any flow may surface supplementary content (e.g., "What to expect during GeneXpert testing" attached to a TB screening referral).

13.3.3 Google AI Interfaces

- **Location normalisation** (all flows)
- **Advisory content tailoring** (personalisation, summarisation, translation)
- **Clinical triage assist** Health — suggest presumptive TB score from symptoms
- **Translation** (English ↔ Swahili; future local languages)
- **Intent extraction** (future)

13.4 Kuza Platform

- BAP and BPP connect to MySQL for all data operations
- Provider and facility catalogs, orders, payments, feedback, conversation state — all persisted in MySQL

13.5 Health-System Integrations

System	Purpose	Mechanism
Disease-Programme Registers (e.g., Kenya National TB Programme)	Read/write to existing registers (for the TB vertical: DSTB, DRTB, TPT, ACF, PTLD and related)	Patient UUID linkage; bidirectional sync per Record step
DHIS2	Aggregated health indicators for national reporting	Scheduled export (monthly) of aggregated counts; no patient-level data
Global Fund / TaifaCare	Outcome reporting for funder visibility	Claim & outcome events emitted on Record + Outcome steps
WHO / CDC	Aggregated surveillance intelligence	Aggregated only; no patient data leaves Kenya
FHIR R4 (optional)	Health data exchange standard for cross-system interop	FHIR resources on demand (Patient, Observation, DiagnosticReport, MedicationDispense)

14. Sovereign Data & Compliance

DATA SOVEREIGNTY

All seeker and patient data is stored **in country (Kenya)**. No patient-identifiable data leaves Kenya. External reporting to WHO / CDC / Global Fund uses aggregated, de-identified indicators only.

Area	Controls
Consent	DPDP-style explicit consent captured before any data is shared with a provider / facility; consent revocable at any time.
Health privacy	HIPAA-equivalent privacy controls; role-based access; facility-scoped visibility; PHI encryption at rest and in transit.
Clinical alignment	All health flows aligned to Kenya Ministry of Health guidelines and the National Disease Programme protocols.
Funder compliance	Outcome reporting structured to Global Fund M&E requirements.
Audit trail	Immutable log of every clinical and financial event, signed and time-stamped, retained per national records policy.
Right-to-erasure	Seekers can request erasure; statutorily retained clinical records are redacted rather than deleted.

15. Non-Functional Requirements

Requirement	Specification
No shopping cart	Each order is a single service/product transaction. No multi-item baskets.
No shipping	Services delivered on-site, at provider location, or at facility.
No online patient payments	Local-services payments offline. Health-services are free to patients.
No app installation	Seekers on WhatsApp only. Providers / facilities / coordinators use WhatsApp + browser web app.
Offline-friendly	Critical flows work on low bandwidth. Minimal data transfer per interaction.
Multilingual	English + Swahili at minimum; additional local languages in future.
Data privacy	See § 14. Seeker PII minimised; health PHI encrypted; cross-border transfer prohibited.
Availability	99% uptime during pilot; WhatsApp webhook (Meta) SLA.
Response time	WhatsApp replies within 5 seconds; provider/facility discovery within 10 seconds.
Clinical safety	Health flows have human-in-the-loop gates — AI triage never issues clinical diagnoses; all results delivered by credentialed clinicians.

16. Pilot Scope & Future Extensions

In Scope for Pilot

#	Feature	Domain	Status
1	All four local-services flows (Livelihood, Household, Market, Advisory)	Local	Required
2	All four health-services flows (TB Screening, Treatment, Supervision, Post-TB)	Health	Required
3	Unified 7-step Flow Engine	Both	Required
4	Domain detection and routing	Both	Required
5	Provider / facility onboarding and catalog management via web app	Both	Required
6	Full lifecycle for transactional flows	Both	Required
7	Advisory + Health Info content delivery	Both	Required
8	Multi-channel access: WhatsApp + browser web app (SMS/USSD optional)	Both	Required
9	Location-based matching with capability filters	Both	Required
10	Disease-programme register integration	Health	Required
11	Phase 1 multi-domain analytics dashboards (see Appendix)	Both	Required
12	Kenya deployment (counties, KES, M-Pesa)	Both	Required

Future Extensions (NOT in Pilot)

Extension	Description
Additional health verticals	HIV, malaria, maternal/child health, NCDs — same flow engine, same 4 generic schemas, new `verticalAttributes.vertical` values.
Additional local-services categories	Financial services, legal, government schemes.

Extension	Description
Cross-domain seeker journeys	Unified view of a seeker who uses both local and health services (with consent).
Phase 2 predictive ML models	Risk stratification, treatment failure prediction, demand forecasting (see Appendix).
Phase 3 longitudinal surveillance	PTLD registry, multi-disease platform, cross-domain network analytics (see Appendix).
Buyer-initiated Market discovery	External buyers seeking produce via aggregators.
Online M-Pesa payment processing	Integrated payment gateway with real-time confirmation.
Provider / Coordinator WhatsApp conversational menu	Catalog and supervision management via WhatsApp (no web app needed).
FHIR R4 full conformance	Complete FHIR resource surface for external EMR interop.
Multi-language AI advisory	AI-powered content delivery in additional local languages.

17. Appendix — Phased Analytics Roadmap

> **ROADMAP, NOT ARCHITECTURE**
>
> This is a **roadmap framework**, not an architectural component. Analytics is implemented as a **logical capability** that consumes data from the platform — it is not a separate layer of the stack. The architecture has 6 layers plus external integrations; analytics sits on top of them as a consumer.

Analytics maturity progresses through three phases regardless of domain. The same phases apply to local services, health services, and any future vertical added to the platform.

17.1 Phase 1 — Analytics on Existing Data (Months 1–6)

Use what you already have. No new data collection required — dashboards and analyses sit on top of the operational data captured by the flow engine.

Domain	Illustrative Dashboards & Analyses
Local	Provider performance dashboards, conversion funnels (discover → select → confirm → deliver), seasonal demand patterns, geographic service-availability heatmaps, cross-category provider utilisation.
Health	Facility performance (presumptive-to-confirmed ratio), screening cascade analysis, supervision-visit coverage gaps, financial anomaly detection on payment claims, CHP productivity.

17.2 Phase 2 — Predictive ML (Months 6–18)

Train models on accumulated outcome data once enough longitudinal history has accrued.

Domain	Illustrative Models
Local	Demand forecasting, provider-matching optimisation, seeker churn prediction, price-signal detection.
Health	Patient risk stratification, treatment failure prediction, contact-investigation prioritisation, mortality prevention (NLP on death audits), ADR early warning.

17.3 Phase 3 — Pioneering Surveillance (Months 12–36)

Longitudinal, multi-domain, novel intelligence — the phase that produces datasets and insights not available elsewhere.

- Local **Cross-domain network effects** — seekers active in multiple verticals show materially different retention and outcome profiles.
- Local **Multi-vertical seeker journeys** — unified view across all verticals a seeker participates in.
- Health **Longitudinal post-TB lung disease surveillance** — national-scale dataset on PTLD trajectories.
- Health **Multi-disease platform extension** — HIV, malaria, cholera on the same flow engine and taxonomy.

> **APPLIES TO ANY DOMAIN**
>
> The three-phase progression — *use what you have → predict → pioneer* — is domain-agnostic. Every vertical added to the platform (health, agri, finance, education, etc.) inherits the same roadmap. The examples above are illustrative, not prescriptive.